

Orbital Geometry Guide

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May 27, 2025

Abstract

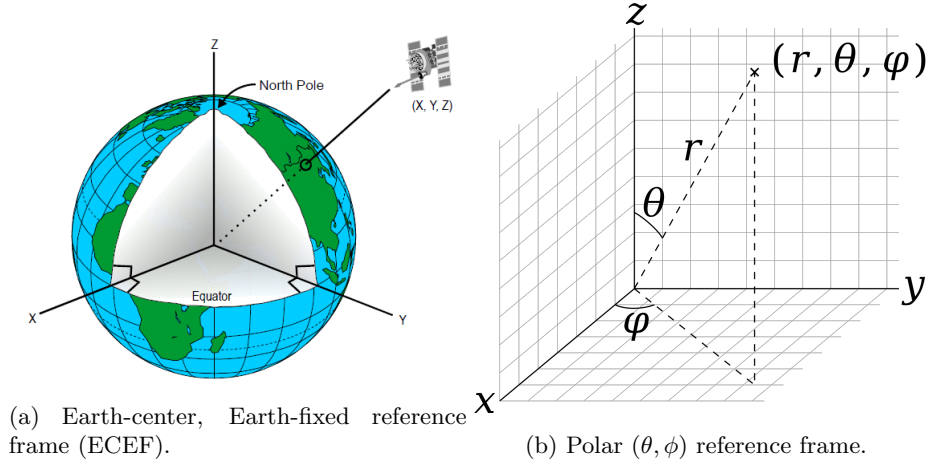


Figure 1: Comparison of the ECEF and Polar reference frames.

Coordinate Systems

ECEF, Polar, & Latitude-Longitude Coordinates

In Figure 1 (a) of the **ECEF** reference frame, the z^+ -axis is the North Pole, x^+ passes through the intersection of the Equator (0° latitude) and Prime Meridian (0° longitude), and the y^+ -axis passes through the Equator and 90° longitude arc.

Figure 1 (b) shows us that in the **polar** coordinate system, θ is the angle measured down from the z^+ -axis toward the xy -plane, and ϕ is the angle measured from the x^+ -axis, counterclockwise.

The only difference between the polar coordinate system and the **latitude-longitude** coordinate system is that a point's latitude is measured up from the xy -plane towards the z^+ axis. In order to find the **altitude** we simply take

$$\text{Alt} = r - r_e$$

, where r_e is the radius of the Earth at that point. (Note: the Earth is elliptical, but for the sake of the calculations in this guide, we will be assuming a spherical Earth with radius $r_e = 6378$ km.)

The conversions between these frames is as follows:

Polar-to-LLA

$$\text{Latitude} = \Theta = 90 - \theta$$

$$\text{Longitude} = \Phi = \phi$$

ECEF-to-LLA (and Polar!)

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\text{Alt} = r - r_e$$

$$\theta = \cos^{-1}\left(\frac{z}{r}\right)$$

$$\text{Latitude} = 90^\circ - \cos^{-1}\left(\frac{z}{r}\right)$$

$$\text{Longitude} = \phi = \tan^{-1}\left(\frac{y}{x}\right)$$

LLA-to-ECEF

$$x = r \cos(\text{Latitude}) \cos(\text{Longitude})$$

$$y = r \cos(\text{Latitude}) \sin(\text{Longitude})$$

$$z = r \sin(\text{Latitude})$$

Polar-to-ECEF

$$x = r \sin(\theta) \cos(\phi)$$

$$y = r \sin(\theta) \sin(\phi)$$

$$z = r \cos(\theta)$$

Platform-centered Coordinate Systems (PCCS)

The local coordinate system of the platforms has the **nadir** vector (that is, the vector pointing from the center of the platform to the center of Earth) as the z^+ -axis. The sub-platform point (SPP) is the point at which the nadir line intersects the surface of the Earth. In the PCCSs, the x^+ -axis parallel to, and directionally the same, as the northward vector of the SPP. For our purposes, the directionality of the y -axis is ambiguous.

Line of Sight (LOS)

First, in order to determine if two platforms have LOS, we need to calculate the λ_0 (the angular region of Earth) seen by each platform.

Clearly, we can see that λ_0 for each platform is given by

$$\lambda_{01} = \frac{r_e}{r_e + alt_1}$$

$$\lambda_{02} = \frac{r_e}{r_e + alt_2}$$

Note that for a platform on the Earth's surface, $\lambda_0 = 0$. Then we need to find λ , which is the inner-Earth angle between the platforms. We can accomplish this by leveraging the Spherical Law of Cosines, which states

In the case of platforms on or orbiting Earth, we can leverage this by looking at the latitude and longitude of the SPPs. These projections will have the same angular relationships as the platforms themselves. So,

The Spherical Law of Cosines becomes

$$\cos \lambda = \cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2 \cos(|\phi_1 - \phi_2|)$$

or in terms of latitude and longitude:

$$\cos \lambda = \sin lat_1 \sin lat_2 + \cos lat_1 \cos lat_2 \cos \Delta L$$

where $\Delta L = |lon_1 - lon_2|$.

Now, we can compare the λ_0 s to λ . If $\lambda \leq \lambda_{01} + \lambda_{02}$, then there is LOS. Otherwise, there is no LOS between the platforms.

As an alternative, we can use the Definition of the Inner Product and the ECEF coordinates found for each platform in order to calculate λ . We can think of the ECEF coordinates as the directional vector for each platform, originating at the center of the Earth. Therefore, using the Dot Product, we observe

$$\vec{p}_1 = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$$

$$\vec{p}_2 = \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$$

$$\mathbf{p}_1 \cdot \mathbf{p}_2 = \sum_{i=1}^n p_{1i} p_{2i} = \|\mathbf{p}_1\| \|\mathbf{p}_2\| \cos \lambda$$

which we can rearrange to

$$\cos \lambda = \frac{x_1 x_2 + y_1 y_2 + z_1 z_2}{\sqrt{x_1^2 + y_1^2 + z_1^2} \sqrt{x_2^2 + y_2^2 + z_2^2}}$$

Another factor in LOS is elevation, ϵ , of the line-of-sight above Earth's surface (not to be confused with the local **azimuth and elevation** "look" angles,

which we will discuss later!) Many communication systems have a minimum elevation angle at which they can reliably operate.

With respect to (w.r.t) platform (1), we see that ρ and λ_0 are related by

$$\cos \lambda_0 = \sin \rho = \frac{r_e}{r_e + alt_1}$$

(Note: if we were calculating w.r.t platform (2), we would use alt_2 instead of alt_1 in this calculation.)

We can calculate η as follows:

We can see from this configuration that

$$\sin \lambda = \frac{a}{r_e + alt_2} \implies a = (r_e + alt_2) \sin \lambda$$

$$\cos \lambda = \frac{b}{r_e + alt_2} \implies b = (r_e + alt_2) \cos \lambda$$

$$(r_e + r_1) - b = (r_e + r_1) - (r_e + alt_2) \cos \lambda$$

we can now use these relationships to determine

$$\tan \eta = \frac{(r_e + alt_2) \sin \lambda}{(r_e + r_1) - (r_e + alt_2) \cos \lambda}$$

Note, that is platform (2) is on the surface of the Earth, $r_e + alt_2 = r_e$, as $alt_2 = 0$, therefore

$$\tan \eta = \frac{r_e \sin \lambda}{(r_e + r_1) - r_e \cos \lambda} \left(\frac{1/r_e + r_1}{1/r_e + r_1} \right)$$

$$\tan \eta = \frac{\sin \rho \sin \lambda}{1 - \sin \rho \cos \lambda}$$

Moreover, we can also calculate η using the Law of Sines. Observe that

$$\frac{\sin \eta_1}{re + alt_2} = \frac{\sin \eta_2}{re + alt_1} = \frac{\sin \lambda}{D}$$

where D is the range between the platforms. Rearranging, we get the following relationships

$$\sin \eta_1 = \frac{(re + alt_2) \sin \lambda}{D}$$

$$\sin \eta_2 = \frac{(re + alt_1) \sin \lambda}{D}$$

$$\eta_2 = 180^\circ - \lambda - \eta_1$$

We can calculate D using the ECEF coordinates of each platforms:

$$D = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

From the diagram, we can tell that we have 2 cases.

(1) if $\rho > \epsilon$ (i.e., for a platform on Earth's surface),

$$\epsilon = 90^\circ - (\eta + \lambda)$$

or, (2) if $\eta \geq \rho$ (i.e., for 2 airborne platforms),

$$\epsilon = \rho - \eta$$

More on LOS

Observe that we can derive the following relationships between η , λ , and ϵ from the above diagram

$$\cos \epsilon = \frac{\sin \eta}{\sin \rho} \implies \sin \eta = \frac{\cos \epsilon}{\sin \rho}$$

and the range between the two platforms, D , can be calculated from

$$D = r_e \frac{\sin \lambda}{\sin \eta}$$

or, more readily, using the ECEF coordinate frame

$$D = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

Now, clearly we will have the range early in our calculations. So, if we are given the minimum above Earth elevation angle, ϵ_{min} , we can calculate the maximum distance between platforms where they will still have LOS to one another, D_{max} , unobstructed by Earth. Observe:

$$\sin \eta_{max} = \sin \rho \cos \epsilon_{min}$$

$$\lambda_{max} = 90^\circ - \epsilon_{min} - \eta_{max}$$

$$D_{max} = r_e \left(\frac{\sin \lambda_{max}}{\sin \eta_{max}} \right)$$

And clearly, if $D \leq D_{max}$, the platforms have LOS.

This may reduce the number of calculations needed to find the necessary angles, given your needs!

Local Viewing Angles

Cardinal Direction	Azimuth
North	0°
South	90°
East	180°
West	270°